Supplement 4: Empirical application, Danish Neolithic settlement data

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Aim

Supplements 1–3 examined the problem in simulations. Here we turn to an empirical dataset, Danish Neolithic settlement records.

The aim is methodological, and we make no claim about the Danish Neolithic itself. The question is:

> How strongly does an empirical aoristic reconstruction depend on the chronological structure of the dating intervals, and how does phase correction change the resulting temporal pattern?

The true generating process for these data is unknown, so there is no ground truth to test the methods against. We therefore compare the following six elements:

1. the structure of the chronological intervals;
2. the standard proportional aoristic reconstruction;
3. the Monte Carlo aoristic reconstruction;
4. the phase-corrected aoristic reconstruction;
5. a Bayesian reconstruction with Baorista (Crema 2025), the model-based benchmark already used in Supplements 2 and 3, based on the same chronological intervals;
6. an external summed calibration of Danish radiocarbon dates for settlements from the XRONOS database (Roe et al. 2025).

In particular, we ask whether peaks, troughs and trends of the reconstruction follow the structure of the chronological classification.

The radiocarbon dates are no ground truth. They cover only part of the sites, have their own taphonomic, research-historical and sampling biases, and summed probability distributions

(SPDs) are a problematic summary of them. Their biases, however, differ from those of the phase system, so agreement with an SPD is informative about the mechanism examined here.

Computation

The Baorista fits for the four priors are computed with `scripts/run_s4_baorista_sensitivity.sh` (`make refresh-s4-baorista`), and this document reads their posterior summaries from `outputs/`. The XRONOS data are read from `data/`. The full reproduction workflow is described in the README.

Packages

```
library(dplyr)
library(tidyr)
library(ggplot2)
library(purrr)
library(tibble)
library(readr)
library(rlang)
library(here)
library(xronos)
library(rcarbon)
```

Settings

```
set.seed(42)
options(knitr.kable.NA = "-")

dir.create("outputs", showWarnings = FALSE, recursive = TRUE)
dir.create("figures", showWarnings = FALSE, recursive = TRUE)
dir.create(here("data"), showWarnings = FALSE, recursive = TRUE)

source("R/aoristic_helpers.R")

input_file <- "../data/anlaeg_settlements.csv"
from_col <- "fra_aar"
to_col <- "til_aar"
```

```r
time_min <- -4100L
time_max <- -1800L
plot_time_min <- -3800L
plot_time_max <- time_max

year_grid <- seq(time_min, time_max - 1, by = 1)

bin_width <- 50L
n_mc <- 100L

inclusive_end_fix <- TRUE
harmonise_free_entries <- Sys.getenv("HARMONISE_FREE_ENTRIES", "true") ==
 ↪  "true"

ff_period_grid <- c(
  -5400L, -4800L, -4300L, -3950L, -3700L, -3500L, -3300L, -3200L, -3100L,
  -3000L, -2900L, -2800L, -2600L, -2450L, -2350L, -1950L, -1700L
)

max_interval_width <- Inf

refresh_xronos <- Sys.getenv("REFRESH_XRONOS", "false") == "true"
xronos_site_type_filter <- "settlement"

xronos_raw_file <- here("data", "xronos_denmark_raw.csv")
xronos_dedup_file <- here("data", "xronos_denmark_c14_dedup.csv")
xronos_report_file <- here("data", "xronos_denmark_dedup_report.csv")
xronos_spd_file <- here("data", "xronos_denmark_sumcal.csv")
xronos_cal_file <- here("data", "xronos_denmark_calibrated.rds")
xronos_spd_rds_file <- here("data", "xronos_denmark_spd.rds")

baorista_rates <- c(1, 10, 50, 100)

baorista_reference_rate <- 1

baorista_file <- function(rate, suffix) {
  file.path("outputs",
            paste0("s4_danish_neolithic_baorista_rate_", rate, "_", suffix))
}

baorista_curve_files <- setNames(
  vapply(baorista_rates, baorista_file, suffix = "curve_binned.csv",
```

```
        FUN.VALUE = character(1)),
  baorista_rates
)

baorista_posterior_files <- setNames(
  vapply(baorista_rates, baorista_file, suffix = "posterior_binned.csv",
        FUN.VALUE = character(1)),
  baorista_rates
)

baorista_diagnostics_files <- setNames(
  vapply(baorista_rates, baorista_file, suffix = "diagnostics.csv",
        FUN.VALUE = character(1)),
  baorista_rates
)

reference_key <- as.character(baorista_reference_rate)
baorista_curve_file <- baorista_curve_files[[reference_key]]
baorista_posterior_file <- baorista_posterior_files[[reference_key]]
baorista_diagnostics_file <- baorista_diagnostics_files[[reference_key]]

xronos_time_range_bp <- c(6500, 2500)

method_colours <- c("Standard aoristic" = "#D55E00",
                    "Monte Carlo mean" = "#0072B2",
                    "Phase-corrected" = "#009E73",
                    "Baorista posterior median" = "#CC79A7")

method_linetypes <- c("Standard aoristic" = "solid",
                      "Monte Carlo mean" = "longdash",
                      "Phase-corrected" = "solid",
                      "Baorista posterior median" = "dotdash")
```

Input data

This supplement uses an empirical dataset of Danish Neolithic settlements. The data derive from the Danish national register of sites and monuments, *Fund og Fortidsminder* (Slots- og Kulturstyrelsen 2018, data downloaded 5 April 2018). The full export (`data/anlaeg_all.csv`) is reduced to Neolithic settlement records by `scripts/prepare_s4_danish_settlements.R`, which writes `data/anlaeg_settlements.csv`, the input of this supplement. For each observation, a beginning and an end date are given. Dates before the common era are represented

by negative values.

```
if (!file.exists(input_file)) stop("Run
 ↪  scripts/prepare_s4_danish_settlements.R first.")

raw_data <- read_csv(input_file, show_col_types = FALSE)
```

The dating intervals are taken from `fra_aar` (from) and `til_aar` (to) and treated as half-open, $[from, to)$, as in the simulations. Since the source records most end dates inclusively (an interval ending at `-2801` includes 2801 BCE, the next phase starts at `-2800`), end dates ending in 1 are shifted by one year. Otherwise every phase boundary would leave a one-year gap in the overlap count.

The analytical window runs from 4100 to 1800 BCE. Observations whose dating intervals overlap it are kept and clipped to it, a compromise for the methodological purpose of this supplement. All filtering, reconstruction, normalisation, and numerical diagnostics use this full window. The only exception is the comparison with the radiocarbon record, which is restricted to the common Neolithic window explained there. Some plots use a narrower display window beginning at 3800 BCE, to keep the low-density edge of the curves from dominating the figure, which affects presentation only.

Data preparation

The Neolithic settlement records were already selected when the input file was prepared (see above), so no further records are removed here. Two steps prepare the dating intervals.

First, free entries are harmonised. Most dates in Fund og Fortidsminder are chosen from the register's period list, which records end years inclusively (e.g. `-2801`) and allows a record to be dated from one period to another. A few records were instead dated freely, for example by a radiocarbon range such as `-3369` to `-3097`. Such ad hoc intervals do not belong to the classification system but would still enter the overlap count $\lambda(t)$. Each of them is therefore assigned to the period of the list, i.e. the combination of two boundaries of the period grid, with which it overlaps most (intersection over union). The assignment is written to `outputs/s4_danish_neolithic_harmonisation_mapping.csv`, and setting `HARMONISE_FREE_ENTRIES=false` restores the original intervals.

```
is_list_entry <- function(end) abs(end) %% 50L == 1L

interval_overlap_ratio <- function(s1, e1, s2, e2) {
  pmax(0, pmin(e1, e2) - pmax(s1, s2)) / (pmax(e1, e2) - pmin(s1, s2))
}
```

```
grid_periods <- expand_grid(grid_start = ff_period_grid,
                            grid_end = ff_period_grid) |>
  filter(grid_end > grid_start)

best_grid_period <- function(start, end) {
  ratio <- interval_overlap_ratio(
    start, end, grid_periods$grid_start, grid_periods$grid_end
  )
  k <- which.max(ratio)

  tibble(grid_start = grid_periods$grid_start[k],
         grid_end = grid_periods$grid_end[k], overlap = ratio[k])
}

raw_dates <- raw_data |>
  mutate(raw_start = as.integer(.data[[from_col]]),
         raw_end = as.integer(.data[[to_col]]),
         list_entry = is_list_entry(raw_end))

free_entries <- raw_dates |>
  filter(!list_entry, is.finite(raw_start), is.finite(raw_end),
         raw_end > raw_start) |>
  count(raw_start, raw_end, name = "n")

harmonisation_mapping <- bind_cols(
  free_entries,
  map2_dfr(free_entries$raw_start, free_entries$raw_end, best_grid_period)
)

write_csv(harmonisation_mapping,
          "outputs/s4_danish_neolithic_harmonisation_mapping.csv")

harmonised_data <- raw_dates |>
  left_join(
    harmonisation_mapping |> select(raw_start, raw_end, grid_start,
↪   grid_end),
    by = c("raw_start", "raw_end")
  ) |>
  mutate(harmonised = harmonise_free_entries & !list_entry &
   ↪  !is.na(grid_start),
         !!from_col := if_else(harmonised, grid_start, raw_start),
         !!to_col := if_else(harmonised, grid_end - 1L, raw_end))
```

Second, the dating intervals are clipped to the analytical time window. The number of retained observations and of harmonised records is given in the data summary below.

```
danish_events <- harmonised_data |>
  mutate(
    start = as.integer(.data[[from_col]]),
    end = as.integer(.data[[to_col]]),
    end = if_else(inclusive_end_fix & (abs(end) %% 10L == 1L), end + 1L, end)
  ) |>
  filter(is.finite(start), is.finite(end), end > start, end > time_min,
         start < time_max) |>
  mutate(start = pmax(start, time_min), end = pmin(end, time_max),
         interval_width = end - start) |>
  filter(end > start, interval_width <= max_interval_width)

if (nrow(danish_events) == 0) {
  stop("No valid intervals remain after clipping to the analysis window.")
}

write_csv(danish_events, "outputs/s4_danish_neolithic_prepared_events.csv")
```

Dating precision varies by a factor of 22 within this catalogue. The narrowest intervals cover 100 years, the widest 2,150, and the median sits at 1,150. Across the retained observations there are 63 distinct intervals in total.

A 100-year interval and a 2,150-year interval enter the sum with the same total weight. The shape of the reconstruction therefore depends on how many intervals of each width happen to exist, and not only on how many sites do.

```
data_summary <- tibble(
  metric = c("Number of records in raw data",
             "Number of records after filtering",
             "Number of records with harmonised free entries",
             "Share of raw records retained", "Earliest retained start",
             "Latest retained end", "Median interval width",
             "Minimum interval width", "Maximum interval width",
             "Number of unique retained intervals"),
  value = c(nrow(raw_data), nrow(danish_events),
   ↪  sum(danish_events$harmonised),
           round(nrow(danish_events) / nrow(raw_data), 3),
```

Table 1: Summary statistics of the Danish Neolithic settlement data.

	Value
Number of records in raw data	3,493
Number of records after filtering	3,493
Number of records with harmonised free entries	43
Earliest retained start	-4,100
Latest retained end	-1,800
Median interval width	1,150
Minimum interval width	100
Maximum interval width	2,150
Number of unique retained intervals	63

```
            min(danish_events$start, na.rm = TRUE),
            max(danish_events$end, na.rm = TRUE),
            median(danish_events$interval_width, na.rm = TRUE),
            min(danish_events$interval_width, na.rm = TRUE),
            max(danish_events$interval_width, na.rm = TRUE),
            n_distinct(paste(danish_events$start, danish_events$end)))
)

write_csv(data_summary, "outputs/s4_danish_neolithic_data_summary.csv")

data_summary |>
  filter(metric != "Share of raw records retained") |>
  mutate(value = format(value, big.mark = ",", trim = TRUE, scientific =
   ↪  FALSE)) |>
  kbl( col.names = c("", "Value"), align = "lr")
```

Empirical chronological interval structure

We begin with the structure of the chronological intervals before turning to the reconstructed curves.

```
empirical_intervals <- danish_events |>
  distinct(start, end) |>
  arrange(start, end) |>
  mutate(interval_id = row_number(), width = end - start)
```

The distribution of intervals is very uneven, and a handful of wide to medium-wide ranges

Table 2: The ten most frequent dating intervals in the Danish Neolithic settlement data.

Rank	Interval (BCE)	Width (years)	Records	Share (%)	Cumulative (%)
1	3950–1800	2150	1271	36.4	36.4
2	3950–2800	1150	734	21.0	57.4
3	2350–1800	550	404	11.6	69.0
4	3950–3300	650	237	6.8	75.8
5	3300–2800	500	188	5.4	81.1
6	2800–2350	450	125	3.6	84.7
7	3300–2600	700	58	1.7	86.4
8	2350–1950	400	48	1.4	87.7
9	2900–2600	300	45	1.3	89.0
10	3000–2800	200	38	1.1	90.1

dominates. 3950–1800 BCE carries 1,271 observations and 3950–2800 BCE a further 734. Between them, these two labels account for 57.4% of the dataset.

```
interval_counts <- danish_events |>
  count(start, end, interval_width, name = "n") |>
  arrange(desc(n), start, end)

write_csv(interval_counts, "outputs/s4_danish_neolithic_interval_counts.csv")

interval_concentration <- interval_counts |>
  arrange(desc(n)) |>
  mutate(rank = row_number(), share = n / sum(n),
         cumulative_share = cumsum(share))

write_csv(interval_concentration,
          "outputs/s4_danish_neolithic_interval_concentration.csv")

interval_concentration |>
  slice_head(n = 10) |>
  transmute(Rank = rank, `Interval (BCE)` = paste0(abs(start), "-",
   ↪  abs(end)),
            `Width (years)` = interval_width, Records = n,
            `Share (%)` = round(100 * share, 1),
            `Cumulative (%)` = round(100 * cumulative_share, 1)) |>
  kbl()
```

```
p_interval_widths <- ggplot(danish_events, aes(x = interval_width)) +
  geom_histogram(bins = 30) +
  labs(title = "Dating precision in the Danish Neolithic dataset",
```

```
       subtitle = "Distribution of chronological interval widths",
       x = "Interval width in years", y = "Number of observations") +
  plot_theme(base_size = 9)

p_interval_widths
```

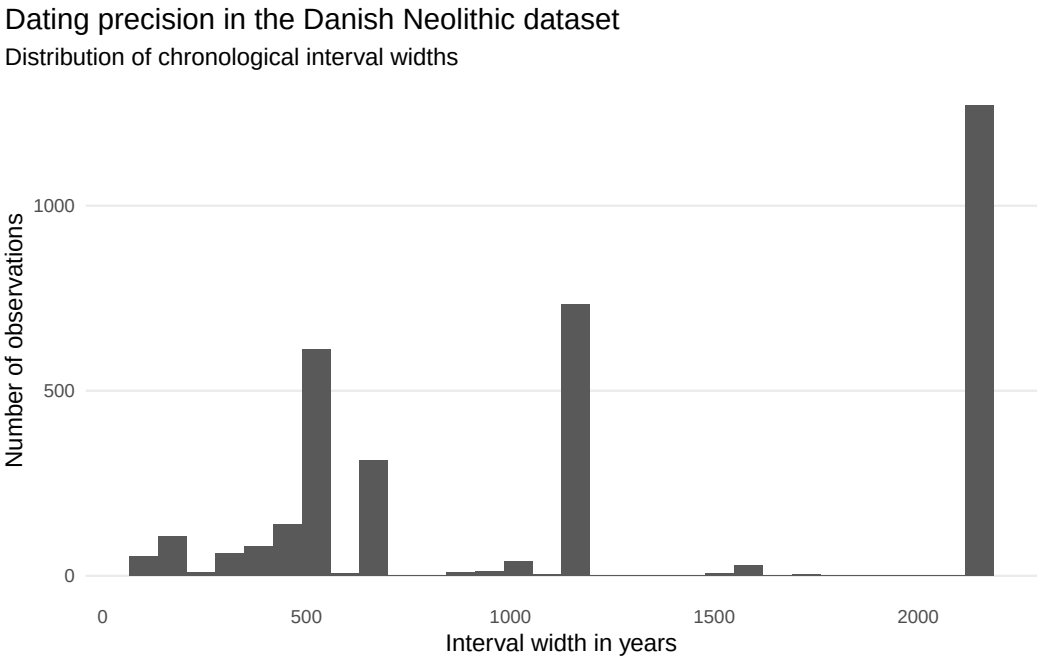


Figure 1: Distribution of the interval widths in the Danish settlement data.

```
p_interval_concentration <- interval_concentration |>
  mutate(interval_label = paste0(start, " to ", end),
         interval_label = reorder(interval_label, n)) |>
  slice_max(n, n = min(25, nrow(interval_concentration))) |>
  ggplot(aes(x = n, y = interval_label)) +
  geom_col() +
  labs(title = "Most frequent chronological intervals in the Danish Neolithic
   ↪  dataset",
       subtitle = "Top empirical interval labels by number of observations",
       x = "Number of observations", y = NULL) +
  plot_theme(base_size = 8)

p_interval_concentration
```

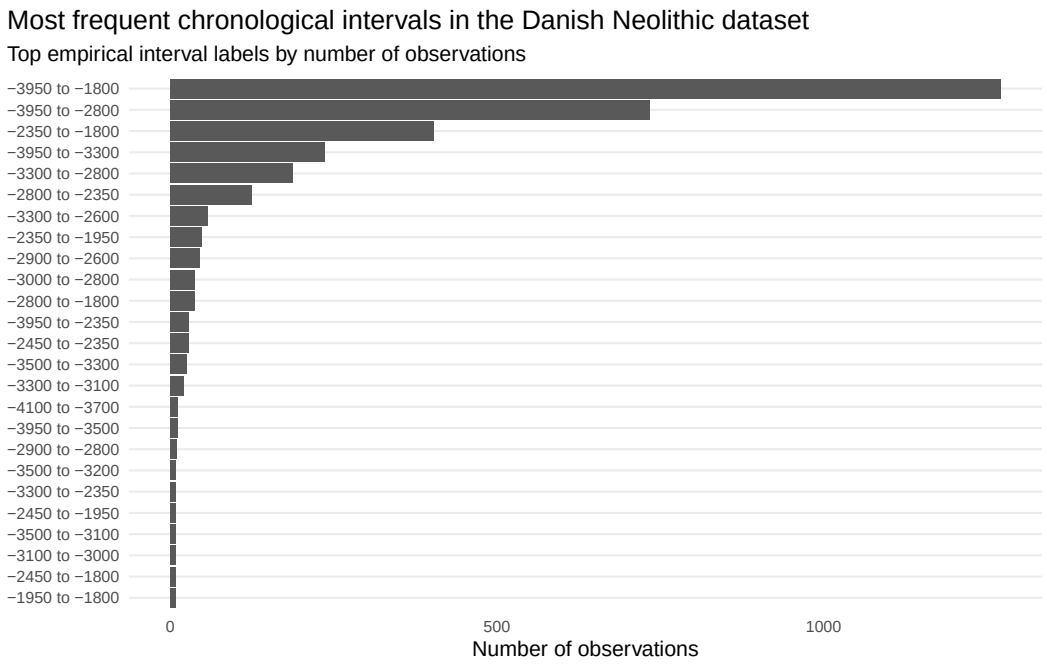


Figure 2: Most frequent dating intervals in the Danish settlement data.

As in the simulations, the data are an uneven mixture of broad and narrow intervals, and the resulting curve may reflect past activity, the structure of the chronological attribution, or both.

Overlap structure

For each year we count the distinct intervals covering it.

```
overlap_curve <- count_phase_overlap(
  phase_catalogue = empirical_intervals |>
    transmute(from = start, to = end),
  year_grid = year_grid
)

overlap_curve_binned <- bin_overlap_curve(
  overlap_curve = overlap_curve, bin_width = bin_width, bin_start = time_min
)
```

```r
p_overlap <- ggplot(overlap_curve_binned, aes(x = year, y = overlap)) +
  geom_line(linewidth = 0.8) +
  geom_point(size = 1.2) +
  coord_cartesian(xlim = c(plot_time_min, plot_time_max)) +
  labs(
    title = "Empirical overlap structure of Danish Neolithic dating
     ↪  intervals",
    subtitle = "Number of unique chronological intervals covering each time
     ↪  bin",
    x = "Year BCE/CE",
    y = "Mean number of overlapping intervals"
  ) +
  plot_theme(base_size = 9)

p_overlap
```

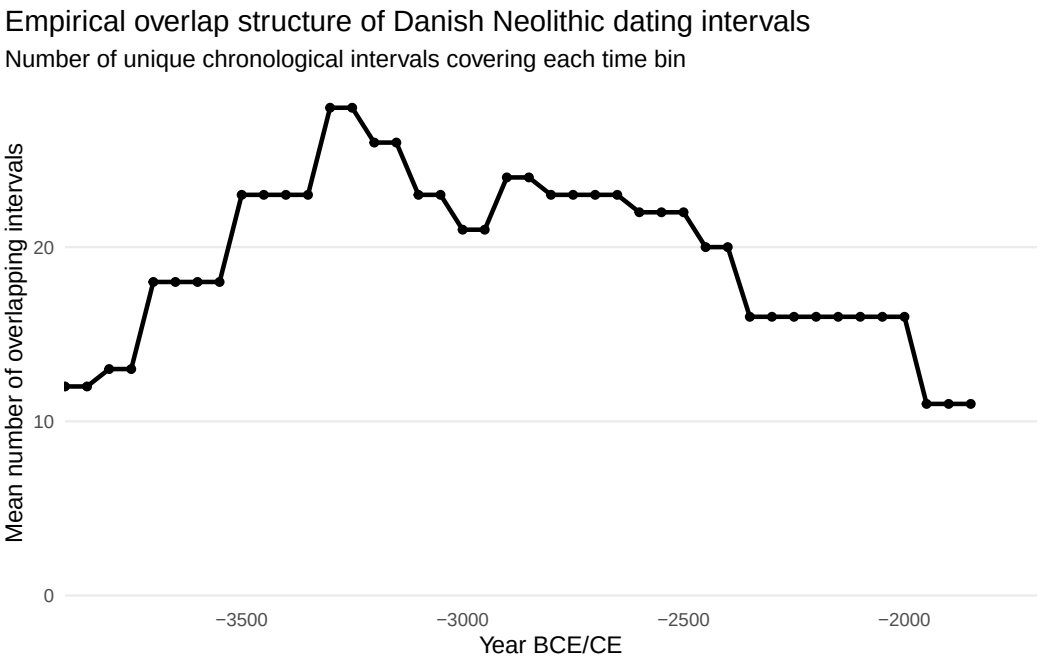


Figure 3: Empirical overlap structure of the Danish dating intervals.

The curve describes only the classification system, that is, how many distinct intervals cover a given time, and says nothing about the number of sites. As shown in the previous supplements, where an empirical reconstruction tracks the overlap curve, part of the reconstructed trajectory may derive from the chronological classification instead from past activity.

Standard proportional aoristic reconstruction

The standard aoristic method distributes each observation uniformly across its assigned chronological interval.

```
standard_curve <- run_standard_aoristic(intervals = danish_events,
                                        year_grid = year_grid)

standard_curve_binned <- standard_curve |>
  make_reconstruction_binned(value_col = standard, output_col = "standard",
                             bin_width = bin_width, bin_start = time_min)
```

For the Danish Neolithic settlement data this produces a rather flat curve with some internal structure and a pronounced shift after 2800 BCE. Many observations carry very wide dating intervals, so a substantial share of the evidence is smeared across long stretches of time. The curve records the distribution of datings as much as the number of observations.

```
p_standard <- ggplot(standard_curve_binned, aes(x = year, y = standard)) +
  geom_line(colour = method_colours[["Standard aoristic"]], linewidth = 0.8,
            na.rm = TRUE) +
  labs(title = "Standard proportional aoristic reconstruction from the Danish
   ↪  Neolithic dataset",
       subtitle = "Binned and normalised to sum to one", x = "Year BCE/CE",
       y = "Normalised intensity") +
  plot_theme(base_size = 9) +
  coord_cartesian(xlim = c(plot_time_min, plot_time_max))

p_standard
```

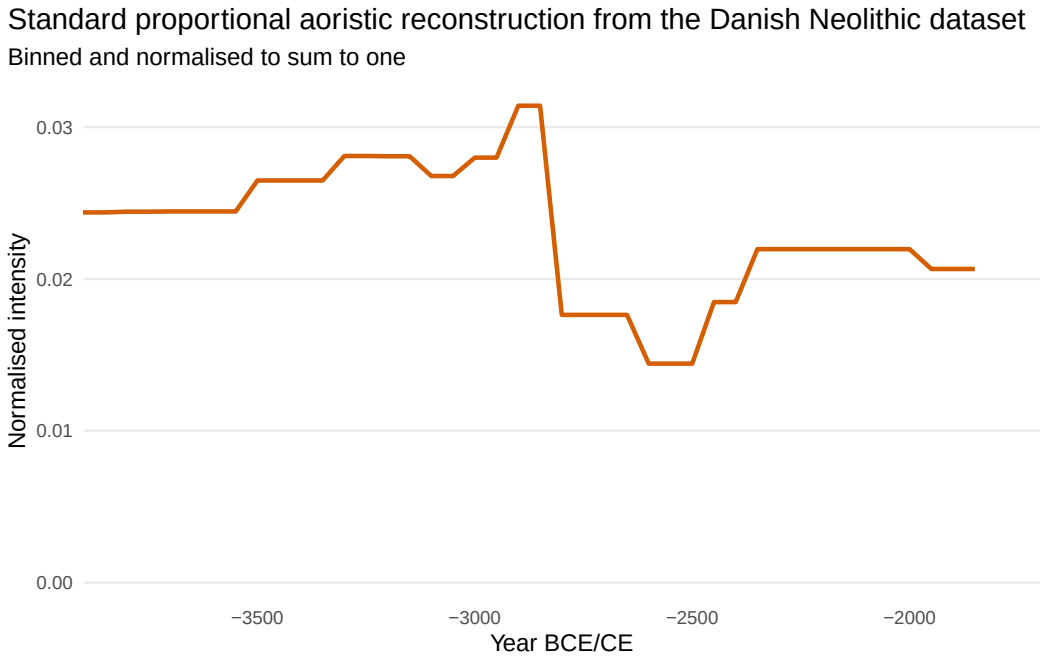


Figure 4: Standard proportional aoristic reconstruction, binned and normalised to sum to one.

Monte Carlo aoristic reconstruction

As in Supplement 2, the Monte Carlo reconstruction repeatedly samples one possible date for each observation from its assigned interval. It expresses uncertainty under the standard within-interval uniformity assumption.

```
mc_curve_binned <- run_monte_carlo_aoristic(
  intervals = danish_events,
  year_grid = year_grid,
  bin_width = bin_width,
  bin_start = time_min,
  n_iter = n_mc,
  seed = 42001
)
```

Setting the uncertainty envelope aside, the Monte Carlo reconstruction is nearly identical to the simple proportional one, as the simulations led us to expect.

```r
p_mc <- ggplot(mc_curve_binned, aes(x = year)) +
  geom_ribbon(aes(ymin = mc_q05, ymax = mc_q95), alpha = 0.20, na.rm = TRUE)
   ↪  +
  geom_line(aes(y = mc_mean), colour = method_colours[["Monte Carlo mean"]],
            linewidth = 0.8, na.rm = TRUE) +
  labs(
    title = "Monte Carlo aoristic uncertainty for the Danish Neolithic
     ↪  dataset",
    subtitle = paste0("Ribbon: 5th to 95th percentile interval across ",
     ↪  n_mc, " iterations"),
    x = "Year BCE/CE",
    y = "Normalised intensity"
  ) +
  plot_theme(base_size = 9) +
  coord_cartesian(xlim = c(plot_time_min, plot_time_max))

p_mc
```

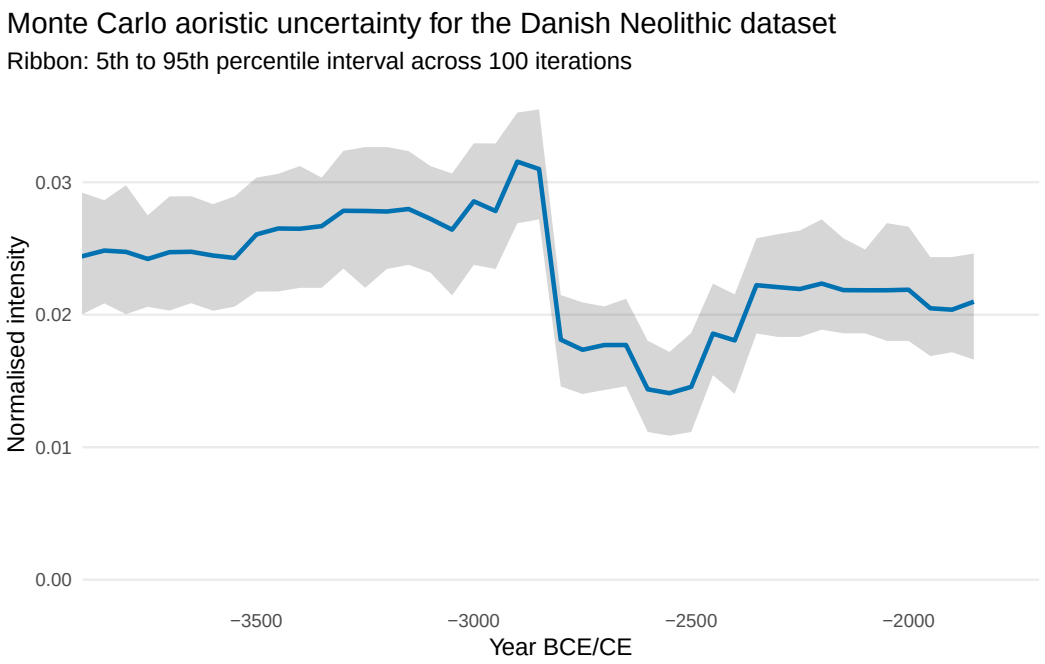


Figure 5: Monte Carlo aoristic reconstruction with the 5th to 95th percentile interval across the iterations.

Phase-corrected aoristic reconstruction

The phase-corrected method reduces the weight assigned to years covered by many overlapping chronological intervals.

```
phase_corrected_curve <- run_phase_corrected_aoristic(
  intervals = danish_events, year_grid = year_grid
)

phase_corrected_curve_binned <- phase_corrected_curve |>
  make_reconstruction_binned(value_col = phase_corrected,
                             output_col = "phase_corrected",
                             bin_width = bin_width, bin_start = time_min)
```

The phase-corrected curve departs substantially from the others. It correlates with the standard aoristic curve at 0.786, so the two are related, but considerable probability is moved between periods. Where the two differ, the interpretation depends on the method.

```
p_phase_corrected <- phase_corrected_curve_binned |>
  inner_join(standard_curve_binned, by = "year") |>
  ggplot(aes(x = year)) +
  geom_line(aes(y = standard), colour = "grey60", linetype = "dashed",
            linewidth = 0.6, na.rm = TRUE) +
  geom_line(aes(y = phase_corrected),
            colour = method_colours[["Phase-corrected"]], linewidth = 0.8,
            na.rm = TRUE) +
  labs(
    title = "Phase-corrected aoristic reconstruction for the Danish Neolithic
     ↪  dataset",
    subtitle = "Binned and normalised to sum to one; dashed grey line:
     ↪  standard reconstruction for reference",
    x = "Year BCE/CE",
    y = "Normalised intensity"
  ) +
  plot_theme(base_size = 9) +
  coord_cartesian(xlim = c(plot_time_min, plot_time_max))

p_phase_corrected
```

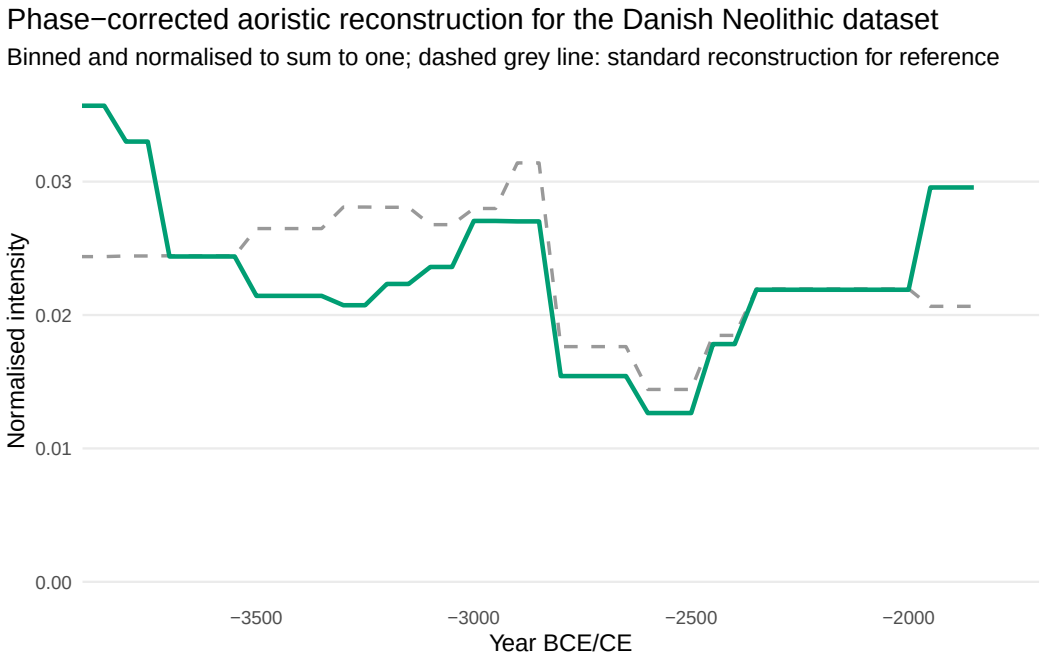


Figure 6: Phase-corrected reconstruction compared with the standard reconstruction (dashed grey line).

Baorista prior-sensitivity analysis

The empirical Baorista reconstruction is evaluated under four exponential priors for the ICAR scale parameter: `dexp(rate=1)`, `dexp(rate=10)`, `dexp(rate=50)`, and `dexp(rate=100)`. All four are fitted to the same 3,493 dated records (63 unique intervals) with the same initial MCMC settings. Fits that do not reach the convergence criterion are refitted with doubled iterations (see the convergence table). This separates features supported by the data from features of the smoothing prior.

The panels below show the posterior mean, since each draw is normalised before summarisation and the mean therefore also sums to one. Ribbons are pointwise 95% credible intervals. The method comparisons further down use the posterior median instead, following the convention of Supplement 3.

```
if (!all(file.exists(c(baorista_curve_files, baorista_posterior_files,
                       baorista_diagnostics_files)))) {
  stop("Run scripts/run_s4_baorista_sensitivity.sh first.")
}

read_baorista_posterior <- function(rate) {
```

```
  read_csv(baorista_posterior_files[[as.character(rate)]],
           show_col_types = FALSE) |>
    mutate(sigma_prior_rate = rate,
           sigma_prior = paste0("dexp(rate=", rate, ")"))
}

read_baorista_diagnostics <- function(rate) {
  read_csv(baorista_diagnostics_files[[as.character(rate)]],
           show_col_types = FALSE) |>
    mutate(sigma_prior_rate = rate)
}

baorista_sensitivity_posterior <- map_dfr(
  baorista_rates, read_baorista_posterior
)

baorista_sensitivity_diagnostics_long <- map_dfr(
  baorista_rates, read_baorista_diagnostics
)

baorista_sensitivity_diagnostics <- baorista_sensitivity_diagnostics_long |>
  filter(
    diagnostic %in% c("max_rhat", "median_rhat", "min_ess", "median_ess",
                      "n_posterior_samples", "niter", "n_attempts",
                       ↪  "converged")
  ) |>
  mutate(value = as.numeric(value)) |>
  select(sigma_prior_rate, diagnostic, value) |>
  pivot_wider(names_from = diagnostic, values_from = value) |>
  arrange(sigma_prior_rate)

baorista_sensitivity_diagnostics |>
  mutate(
    sigma_prior_rate = as.character(sigma_prior_rate),
    across(any_of(c("max_rhat", "median_rhat")),
           ~ formatC(.x, format = "f", digits = 3)),
    across(
      any_of(c("min_ess", "median_ess", "n_posterior_samples", "niter",
       ↪  "n_attempts")),
      ~ format(round(.x), big.mark = ",", scientific = FALSE, trim = TRUE)
    ),
    across(any_of("converged"), ~ if_else(.x == 1, "yes", "no"))
  ) |>
```

Table 3: Convergence diagnostics for the Baorista prior-sensitivity fits.

Prior rate	Max R-hat	Median R-hat	Min ESS	Median ESS	Draws	Iterations	Attempts	Converged
1	1.076	1.002	1,174	2,000	2,000	200,000	3	no
10	1.054	1.002	1,482	2,000	2,000	200,000	3	no
50	1.008	1.002	1,471	2,000	2,000	50,000	1	yes
100	1.008	1.002	1,462	2,000	2,000	50,000	1	yes

```
  select(any_of(c(
    "sigma_prior_rate", "max_rhat", "median_rhat", "min_ess", "median_ess",
    "n_posterior_samples", "niter", "n_attempts", "converged"
  ))) |>
  rename(any_of(c(`Prior rate` = "sigma_prior_rate", `Max R-hat` =
   ↪  "max_rhat",
                  `Median R-hat` = "median_rhat", `Min ESS` = "min_ess",
                  `Median ESS` = "median_ess",
                  `Draws` = "n_posterior_samples",
                  Iterations = "niter", Attempts = "n_attempts",
                  Converged = "converged"))) |>
  (\(d) kbl(
    d,
    align = c("l", rep("r", ncol(d) - 1)))))()
```

```
baorista_sensitivity_plot_data <- baorista_sensitivity_posterior |>
  filter(year >= plot_time_min, year <= plot_time_max) |>
  mutate(
    sigma_prior = factor(sigma_prior,
                         levels = paste0("dexp(rate=", baorista_rates, ")"))
  )

p_baorista_sensitivity <- ggplot(baorista_sensitivity_plot_data,
                                 aes(x = year, y = baorista_mean)) +
  geom_ribbon(aes(ymin = baorista_q025, ymax = baorista_q975), alpha = 0.18)
   ↪  +
  geom_line(linewidth = 0.8) +
  facet_wrap(vars(sigma_prior), ncol = 2, scales = "free_y") +
  labs(title = "Baorista prior-sensitivity analysis",
       subtitle = "Posterior mean and pointwise 95% credible interval",
       x = "Year BCE/CE", y = "Posterior probability") +
  plot_theme(base_size = 9)
```

```
p_baorista_sensitivity
```

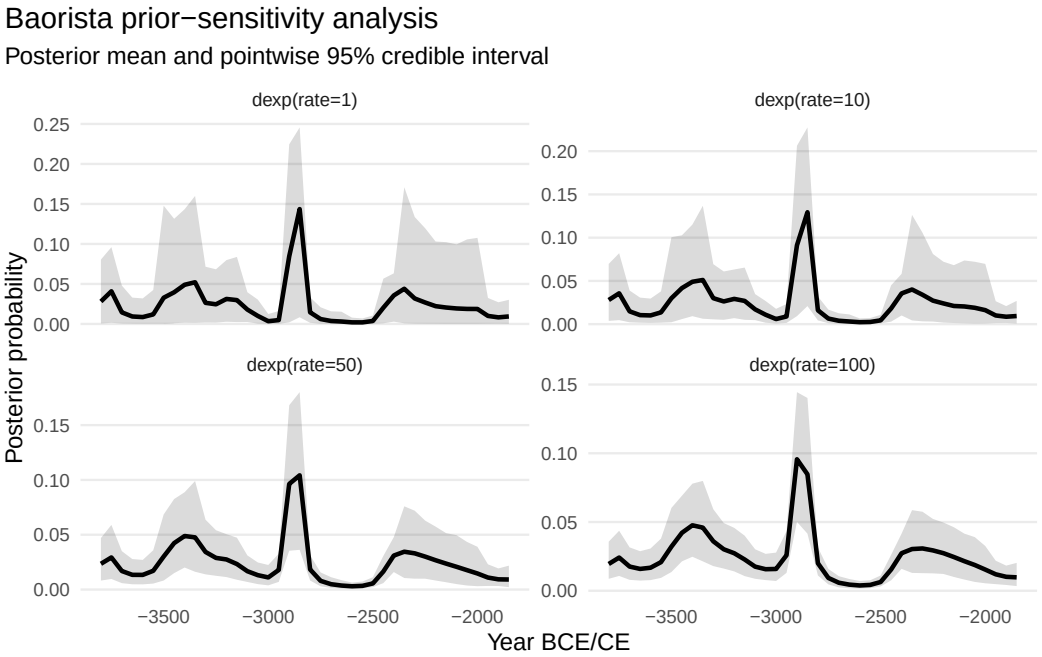


Figure 7: Baorista reconstructions under four priors for the ICAR scale parameter. Posterior mean and pointwise 95% credible interval.

The fits with prior rates 1 and 10 did not reach the convergence criterion after 3 attempts (maximum R-hat 1.08 and 1.05) and are reported but flagged. The peak discussed below appears in all four fits, including the converged ones. The prior affects amplitude and smoothness but leaves the qualitative shape unchanged. As the exponential rate rises, the dominant peak flattens, secondary variation shrinks, and the posterior intervals narrow. Depending on the prior, the maximum of the posterior median falls in the 50-year blocks 2900–2851 BCE and 2850–2801 BCE, i.e. in the century immediately before 2800 BCE. A secondary concentration sits immediately before 3300 BCE, another boundary where frequently repeated ranges end. This is the behaviour expected for interval-censored data. Likelihood-based estimates concentrate probability mass on the intersections of the observed intervals (the Turnbull intervals, Turnbull 1976), and where many intervals share an end point these intersections sit immediately before it.

Because the peak persists across very different priors, it is not caused by the smoothing prior. This does not mean, however, that it reflects past activity. Robustness to the prior and robustness to the chronological observations are two different properties, and only the first has been tested here. All four models condition on the same empirical interval catalogue, and the

peak lies immediately before a major shared boundary at 2800 BCE, where many frequently repeated ranges terminate.

The rate-1 model retains the original Baorista prior and serves as the reference reconstruction in the method comparisons below. It is used only to keep the figures readable.

Baorista posterior reconstruction

For the comparison with the aoristic reconstructions, we use the Baorista fit with prior rate 1 as reference. Its central curve is the posterior median, which handles the asymmetry of these posteriors better than the mean. Its 95% credible interval expresses uncertainty conditional only on the Danish interval data, the Baorista model, and the selected prior.

```
baorista_curve_binned <- read_csv(baorista_curve_file, show_col_types =
 ↪  FALSE)

baorista_posterior_binned <- read_csv(baorista_posterior_file,
                                      show_col_types = FALSE)

baorista_diagnostics <- read_csv(baorista_diagnostics_file,
                                 show_col_types = FALSE)
```

```
curves_binned <- overlap_curve_binned |>
  left_join(standard_curve_binned, by = "year") |>
  left_join(
    mc_curve_binned |>
      select(year, mc_mean, mc_median, mc_q05, mc_q95),
    by = "year"
  ) |>
  left_join(phase_corrected_curve_binned, by = "year") |>
  left_join(
    baorista_posterior_binned |>
      select(year, baorista_median, baorista_q025, baorista_q975),
    by = "year"
  ) |>
  mutate(overlap = replace_na(overlap, 0),
         standard = normalise_curve(replace_na(standard, 0)),
         mc_mean = normalise_curve(replace_na(mc_mean, 0)),
         mc_median = normalise_curve(replace_na(mc_median, 0)),
         phase_corrected = normalise_curve(replace_na(phase_corrected, 0)),
         baorista_median = replace_na(baorista_median, 0),
```

```
         baorista_q025 = replace_na(baorista_q025, 0),
         baorista_q975 = replace_na(baorista_q975, 0))

write_csv(curves_binned,
          "outputs/s4_danish_neolithic_reconstruction_curves_binned.csv")
```

Reconstruction methods and overlap structure

```
overlap_diagnostics <- curves_binned |>
  select(overlap, standard, mc_mean, phase_corrected, baorista_median) |>
  summarise(standard = safe_cor(overlap, standard),
            mc_mean = safe_cor(overlap, mc_mean),
            phase_corrected = safe_cor(overlap, phase_corrected),
            baorista_median = safe_cor(overlap, baorista_median)) |>
  pivot_longer(cols = everything(), names_to = "method",
               values_to = "overlap_curve_correlation") |>
  mutate(
    method = recode(method, standard = "Standard aoristic",
                    mc_mean = "Monte Carlo mean",
                    phase_corrected = "Phase-corrected",
                    baorista_median = "Baorista posterior median")
  ) |>
  arrange(desc(abs(overlap_curve_correlation))) |>
  mutate(
    # overlap-aligned relative amplitude (see Supplement 2)
    overlap_relative_amplitude = map_dbl(
      method,
      ~ overlap_relative_amplitude(
          curves_binned[[c(
      "Standard aoristic" = "standard",
      "Monte Carlo mean" = "mc_mean",
      "Phase-corrected" = "phase_corrected",
      "Baorista posterior median" = "baorista_median"
          )[[.x]]]],
          curves_binned$overlap
        )
    )
  )

write_csv(overlap_diagnostics,
```

Table 4: Danish Neolithic settlement data: correlation of methods with phase-overlap structure.

Method	Correlation with overlap	Overlap-aligned relative amplitude
Standard aoristic	0.661	0.213
Monte Carlo mean	0.656	0.211
Baorista posterior median	0.319	0.477
Phase-corrected	0.158	0.058

```
          "outputs/s4_danish_neolithic_overlap_diagnostics.csv")

overlap_diagnostics |>
  select(method, overlap_curve_correlation, overlap_relative_amplitude) |>
  kbl(
    digits = 3,
    col.names = c("Method", "Correlation with overlap", "Overlap-aligned
     ↪  relative amplitude")
  )
```

```
p_overlap_correlation <- overlap_diagnostics |>
  mutate(method = reorder(method, abs(overlap_curve_correlation))) |>
  ggplot(aes(x = overlap_curve_correlation, y = method)) +
  geom_col() +
  geom_vline(xintercept = 0, linewidth = 0.4) +
  labs(
    title = "Danish Neolithic data: reconstructions and phase overlap",
    subtitle = "Correlation between each method and the empirical overlap
     ↪  curve",
    x = "Correlation with overlap structure",
    y = NULL
  ) +
  plot_theme(base_size = 9)

p_overlap_correlation
```

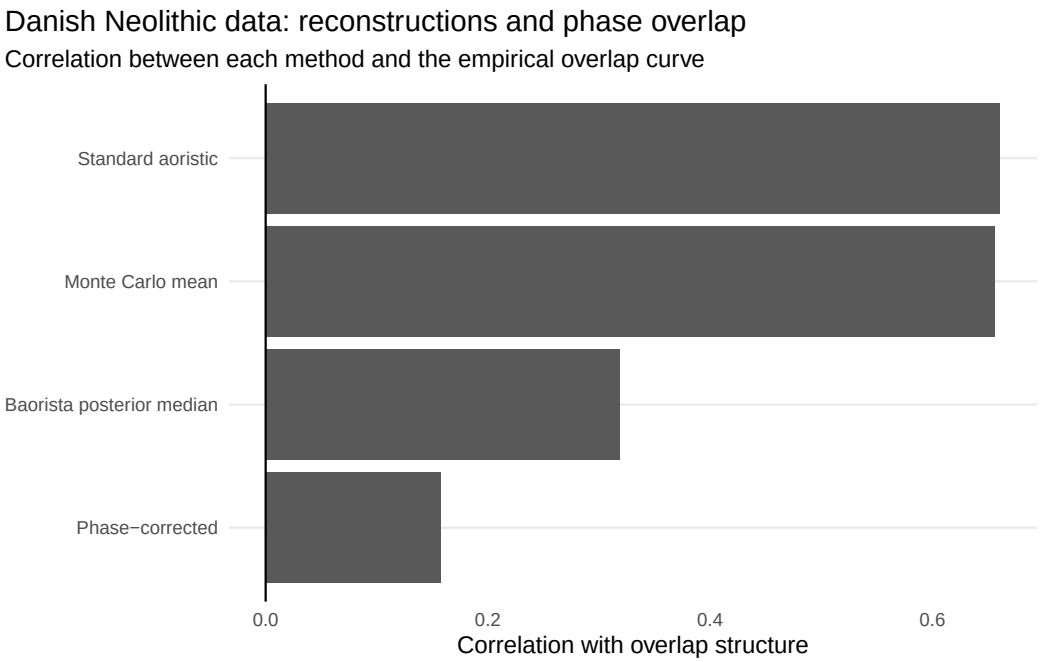


Figure 8: Correlation between each reconstruction and the empirical overlap curve.

The standard proportional result correlates with the empirical overlap curve at 0.661 and the Monte Carlo result at 0.656. The phase-corrected reconstruction drops to 0.158, with the Baorista posterior median between the two at 0.319. Phase correction reduces the coupling most, by a wide margin.

Baorista's lower correlation does not imply weaker coupling. A single coefficient summarises the entire binned series, whereas Baorista's dominant peak is a local feature closely aligned with a major shared interval boundary. Such a local alignment can persist despite a low overall correlation.

```
# The y axis of the comparison figures is cut at 0.045, so that the Baorista
# peak does not flatten the other curves. The peak value is given in the
 ↪  plot.
y_cut <- 0.045

peak_note <- function(curves) {
  peak <- curves |>
    filter(method == "Baorista posterior median") |>
    slice_max(value, n = 1)
  annotate("text", x = peak$year + 40, y = y_cut, hjust = 0, vjust = 1,
           size = 2.8, colour = "#CC79A7",
           label = paste0("Baorista peak ", formatC(peak$value, format = "f",
            ↪  digits = 2),
```

```
                          " (cut off)"))
}
```

```
plot_overlap_methods <- curves_binned |>
  mutate(overlap_scaled = normalise_curve(overlap)) |>
  select(year, Overlap_structure = overlap_scaled, Standard_aoristic =
   ↪  standard,
         Monte_Carlo_mean = mc_mean, Phase_corrected = phase_corrected,
         Baorista_posterior_median = baorista_median) |>
  pivot_longer(cols = -year, names_to = "method", values_to = "value") |>
  mutate(
    method = recode(method, Overlap_structure = "Overlap structure",
                    Standard_aoristic = "Standard aoristic",
                    Monte_Carlo_mean = "Monte Carlo mean",
                    Phase_corrected = "Phase-corrected",
                    Baorista_posterior_median = "Baorista posterior median"),
    method = factor(
      method,
      levels = c("Overlap structure", "Standard aoristic", "Monte Carlo
       ↪  mean",
                 "Phase-corrected", "Baorista posterior median")
    )
  )

overlap_colours <- c("Overlap structure" = "black",
                     "Standard aoristic" = "#D55E00",
                     "Monte Carlo mean" = "#0072B2",
                     "Phase-corrected" = "#009E73",
                     "Baorista posterior median" = "#CC79A7")

overlap_linetypes <- c("Overlap structure" = "dotted",
                       "Standard aoristic" = "solid",
                       "Monte Carlo mean" = "longdash",
                       "Phase-corrected" = "solid",
                       "Baorista posterior median" = "dotdash")

p_overlap_methods <- ggplot(
  plot_overlap_methods,
  aes(x = year, y = value, colour = method, linetype = method)
) +
  geom_ribbon(data = curves_binned |>
                filter(year >= plot_time_min, year <= plot_time_max),
```

```
              aes(x = year, ymin = baorista_q025, ymax = baorista_q975),
              inherit.aes = FALSE, fill = "#CC79A7", alpha = 0.08) +
  geom_line(linewidth = 0.8, na.rm = TRUE) +
  scale_colour_manual(values = overlap_colours, drop = FALSE) +
  scale_linetype_manual(values = overlap_linetypes, drop = FALSE) +
  coord_cartesian(xlim = c(plot_time_min, plot_time_max), ylim = c(0, y_cut))
   ↪  +
  peak_note(plot_overlap_methods) +
  labs(
    title = "Danish Neolithic settlement data: reconstruction methods and
     ↪  empirical phase-overlap structure",
    subtitle = paste0(
      "All curves are normalised for visual comparison; shaded: Baorista 95%
       ↪  credible interval; displayed from ",
      abs(plot_time_min), " to ", abs(plot_time_max), " BCE"
    ),
    x = "Year BCE/CE",
    y = "Normalised value"
  ) +
  plot_theme(base_size = 9)

p_overlap_methods
```

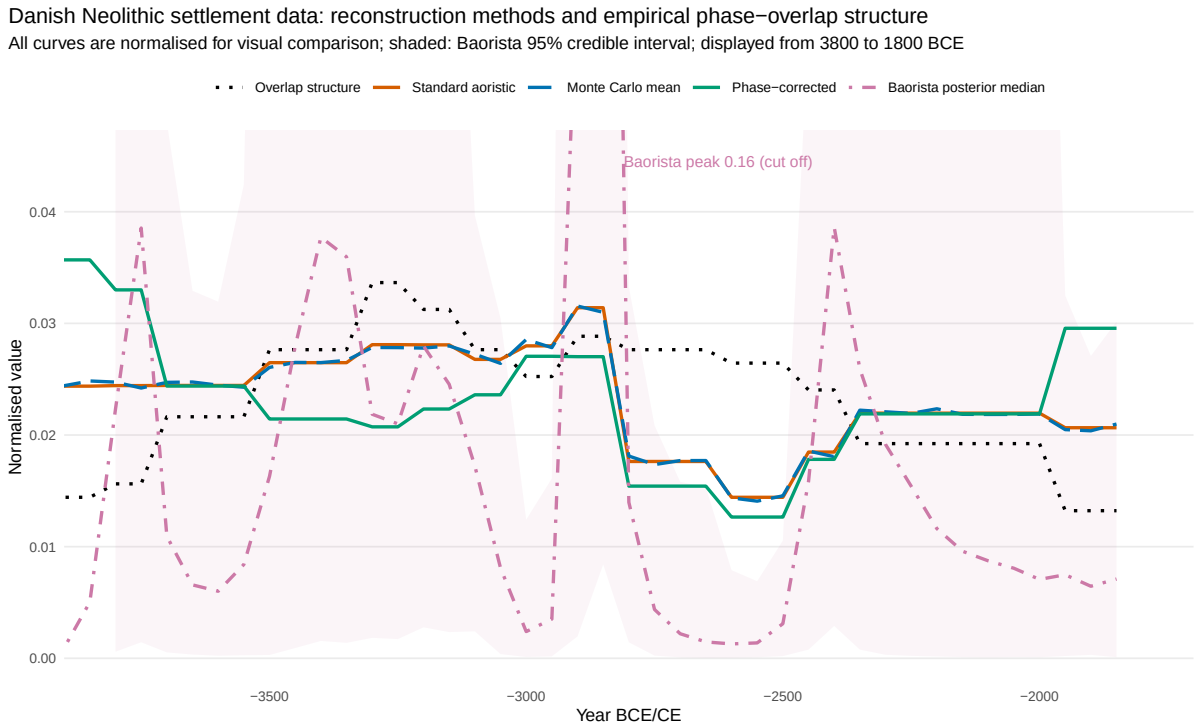


Figure 9: Reconstruction methods and empirical overlap structure, all normalised. The shaded band is the 95% credible interval of the Baorista reconstruction. The vertical axis is cut at 0.045, and the value of the Baorista peak is given in the plot.

Standard and Monte Carlo overlap almost perfectly (Figure 9). The phase-corrected curve shifts the relative weight of several periods while keeping much of the broad trajectory. The Baorista reference fit, summarised by its posterior median, sits well away from all three and is dominated by the peak at 2900–2801 BCE.

The four curves differ most at chronologically structured transitions, especially around the shared interval boundary at 2800 BCE. These sections are sensitive to the combined effect of reconstruction method and chronological classification and should not be read directly as demographic or settlement-historical events.

In the earlier part of the sequence, the phase-corrected curve tends to be lower where overlap is high, in the later part it does not. An exact correction would leave no relationship with the overlap curve, so a negative local relationship reflects either a real signal that runs against the catalogue or overcompensation. The latter is expected where label use is more concentrated than equal choice assumes, as in these data (Supplement 6), and this comparison alone cannot separate the two.

```
ggsave("figures/s4_danish_neolithic_methods_vs_overlap.png",
 ↪  p_overlap_methods,
       width = 8.5, height = 5.2, dpi = 600)
```

Deviation from the overlap structure

The difference between each normalised reconstruction and the normalised overlap curve shows where along the sequence the reconstructions depart from the catalogue (Figure 10). A reconstruction that only reproduced the catalogue would stay close to zero throughout. Because the Baorista peak is much larger than the other deviations, each panel has its own vertical scale.

```
p_deviation <- curves_binned |>
  filter(year >= plot_time_min, year <= plot_time_max) |>
  mutate(overlap_scaled = normalise_curve(overlap)) |>
  transmute(year,
            `Standard aoristic` = standard - overlap_scaled,
            `Monte Carlo mean` = mc_mean - overlap_scaled,
            `Phase-corrected` = phase_corrected - overlap_scaled,
            `Baorista posterior median` = baorista_median - overlap_scaled)
             ↪  |>
  pivot_longer(-year, names_to = "method", values_to = "deviation") |>
  mutate(method = factor(method, levels = names(method_colours))) |>
  ggplot(aes(x = year, y = deviation, colour = method, linetype = method)) +
  geom_hline(yintercept = 0, linewidth = 0.3) +
  geom_line(linewidth = 0.8, na.rm = TRUE, show.legend = FALSE) +
  facet_wrap(vars(method), ncol = 2, scales = "free_y") +
  scale_colour_manual(values = method_colours) +
  scale_linetype_manual(values = method_linetypes) +
  labs(title = "Danish Neolithic data: deviation of the reconstructions from
   ↪  the overlap structure",
       subtitle = "Normalised reconstruction minus normalised overlap curve",
       x = "Year BCE/CE", y = "Difference in normalised density") +
  plot_theme(base_size = 9)

p_deviation
```

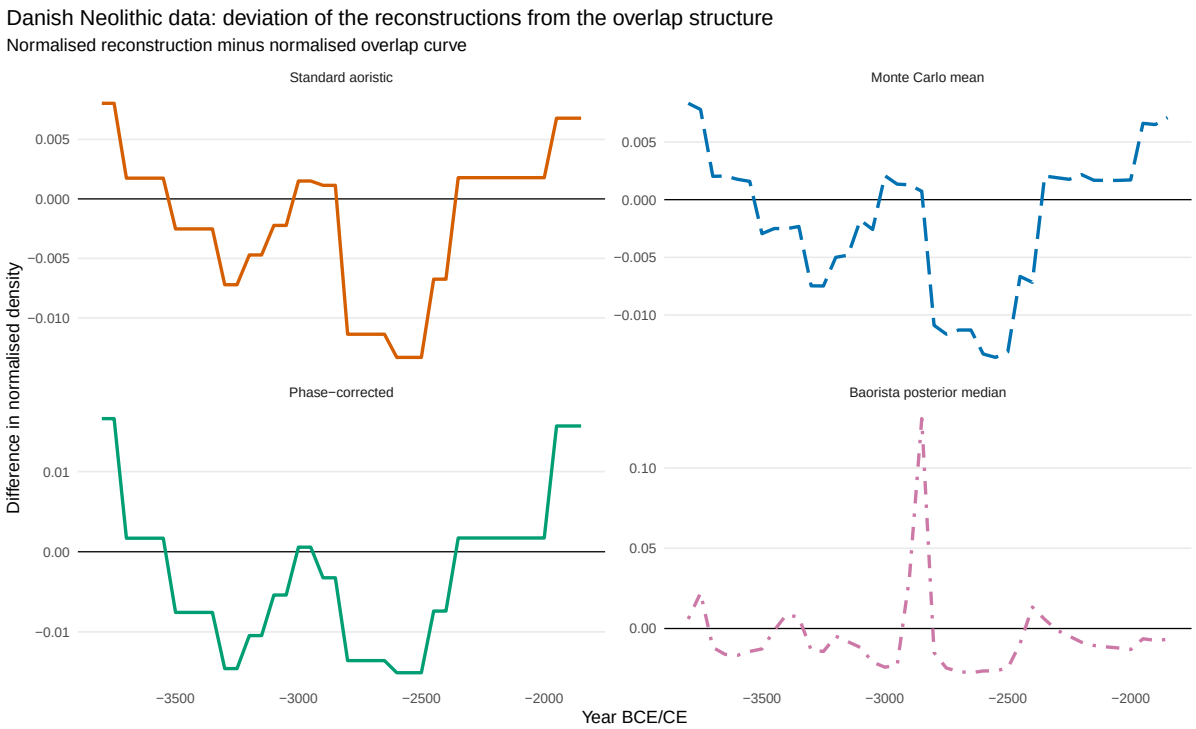


Figure 10: Deviation of the reconstructions from the overlap structure, given as normalised reconstruction minus normalised overlap curve. Each panel has its own vertical scale.

Strength of the overlap coupling

As in Supplement 2, we complement the correlations with the overlap-aligned relative amplitude, $|\beta| \cdot \mathrm{sd}(\lambda)/\bar{R}$, with β the slope of the reconstruction on the overlap curve: the change in reconstructed density, as a fraction of the mean density, when the overlap increases by one standard deviation (see Supplement 2). All curves are normalised to the same total, so the values are comparable across methods.

```
s4_strength <- overlap_diagnostics |>
  select(method, overlap_curve_correlation, overlap_relative_amplitude)

amp <- function(m) s4_strength$overlap_relative_amplitude[s4_strength$method
 ↪  == m]
```

For the standard reconstruction, the density rises by about 21% of the mean per standard deviation of overlap, for the phase-corrected reconstruction by about 6%. For the Baorista

posterior median, correlation alone is misleading. Its correlation with the overlap curve is lower than that of the standard reconstruction (0.319), but because the curve as a whole varies much more strongly, the component aligned with the overlap structure is larger than for the standard reconstruction (48%).

```
p_overlap_scatter <- plot_overlap_scatter(
  curves = curves_binned,
  method_cols = c("standard", "mc_mean", "phase_corrected",
   ↪  "baorista_median"),
  method_labels = c(standard = "Standard aoristic",
                    mc_mean = "Monte Carlo mean",
                    phase_corrected = "Phase-corrected",
                    baorista_median = "Baorista posterior median"),
  colours = method_colours
) +
  labs(title = "Danish Neolithic settlement data: reconstructed density
   ↪  against phase overlap")

p_overlap_scatter
```

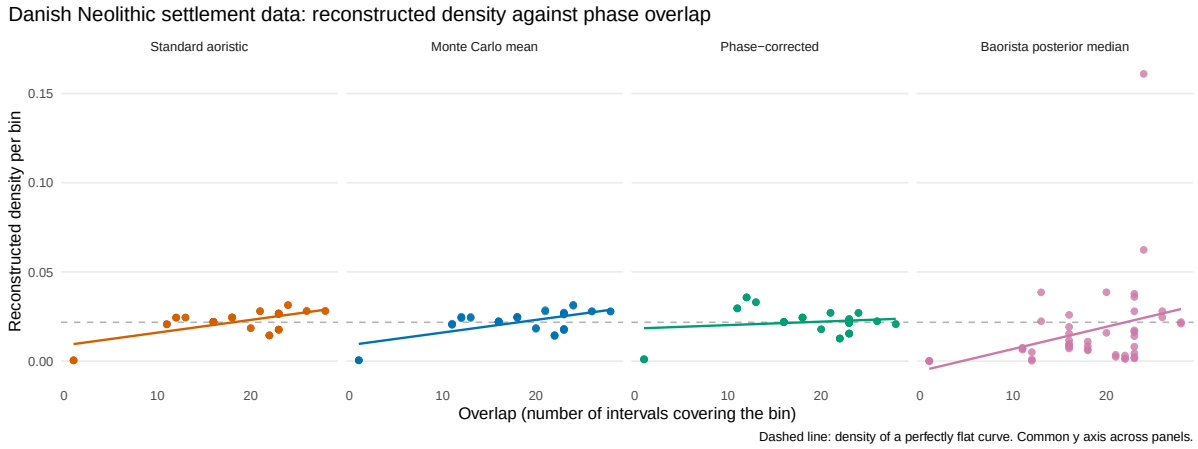


Figure 11: Reconstructed density against phase overlap, by method.

Pairwise method differences

Because there is no known ground truth, we evaluate how strongly the methods differ from one another.

```
method_matrix <- curves_binned |>
  select(standard, mc_mean, phase_corrected, baorista_median)

pairwise_method_correlations <- cor(method_matrix,
                                    use = "pairwise.complete.obs") |>
  as.data.frame() |>
  rownames_to_column("method_1") |>
  pivot_longer(cols = -method_1, names_to = "method_2",
               values_to = "correlation") |>
  mutate(
    method_1 = recode(method_1, standard = "Standard aoristic",
                      mc_mean = "Monte Carlo mean",
                      phase_corrected = "Phase-corrected",
                      baorista_median = "Baorista posterior median"),
    method_2 = recode(method_2, standard = "Standard aoristic",
                      mc_mean = "Monte Carlo mean",
                      phase_corrected = "Phase-corrected",
                      baorista_median = "Baorista posterior median")
  )

write_csv(pairwise_method_correlations,
          "outputs/s4_danish_neolithic_pairwise_method_correlations.csv")
```

```
pairwise_method_distances <- expand_grid(
  method_1 = names(method_matrix), method_2 = names(method_matrix)
) |>
  mutate(
    rmsd = map2_dbl(method_1, method_2,
                    ~ rmsd(method_matrix[[.x]], method_matrix[[.y]])),
    mae = map2_dbl(method_1, method_2,
                   ~ mae(method_matrix[[.x]], method_matrix[[.y]])),
    method_1 = recode(method_1, standard = "Standard aoristic",
                      mc_mean = "Monte Carlo mean",
                      phase_corrected = "Phase-corrected",
                      baorista_median = "Baorista posterior median"),
    method_2 = recode(method_2, standard = "Standard aoristic",
                      mc_mean = "Monte Carlo mean",
                      phase_corrected = "Phase-corrected",
                      baorista_median = "Baorista posterior median")
  )
```

```
pairwise_labels <- c("Standard aoristic" = "Standard",
                     "Monte Carlo mean" = "Monte Carlo",
                     "Phase-corrected" = "Phase-corrected",
                     "Baorista posterior median" = "Baorista")
pairwise_order <- names(pairwise_labels)

pairwise_table <- matrix("-", nrow = length(pairwise_order),
                         ncol = length(pairwise_order),
                         dimnames = list(pairwise_labels, pairwise_labels))

for (i in seq_along(pairwise_order)) {
  for (j in seq_along(pairwise_order)) {
    if (i > j) {
      pairwise_table[i, j] <- formatC(
        pairwise_method_correlations$correlation[
          pairwise_method_correlations$method_1 == pairwise_order[i] &
            pairwise_method_correlations$method_2 == pairwise_order[j]
        ],
        format = "f", digits = 3
      )
    } else if (i < j) {
      pairwise_table[i, j] <- formatC(
        1000 * pairwise_method_distances$rmsd[
          pairwise_method_distances$method_1 == pairwise_order[i] &
            pairwise_method_distances$method_2 == pairwise_order[j]
        ],
        format = "f", digits = 2
      )
    }
  }
}

kbl(pairwise_table, align = "rrrr")
```

The Monte Carlo mean and the standard aoristic result are numerically almost the same curve. The phase-corrected reconstruction stays recognisably related to both, but RMSD shows it moving a substantial amount of probability between periods. Baorista sits 0.024–0.025 RMSD from the aoristic curves, where the three aoristic variants sit within 0.005 of each other. Two defensible inferential frameworks applied to the same interval catalogue thus differ by roughly a factor of 4.8.

Table 5: Danish Neolithic settlement data: pairwise comparison of the reconstructions. Below the diagonal: correlation; above the diagonal: RMSD (×1000) between the binned, normalised curves.

	Standard	Monte Carlo	Phase-corrected	Baorista
Standard	–	0.26	4.97	23.65
Monte Carlo	0.999	–	4.94	23.71
Phase-corrected	0.786	0.788	–	25.36
Baorista	0.424	0.417	0.200	–

Additional comparison: radiocarbon evidence

As an external comparison, we use settlement-related radiocarbon dates from Denmark available through XRONOS (Roe et al. 2025), retrieved with the `xronos` R package on 19 June 2026 and included in the repository. Since the empirical aoristic dataset covers Denmark as a whole, the XRONOS comparison is also extracted at the country level.

To reduce the influence of sites with many radiocarbon determinations, the summed calibration is binned by site name, so that each site contributes as one bin.

XRONOS data and deduplication

```
need_xronos_download <- refresh_xronos || !file.exists(xronos_raw_file)

if (need_xronos_download) {
  message("Downloading Danish chronological data from XRONOS...")

  chron_dk_raw <- chron_data(country = "DK")

  write_csv(chron_dk_raw, xronos_raw_file)
} else {
  message("Using cached raw XRONOS data: ", xronos_raw_file)

  chron_dk_raw <- read_csv(xronos_raw_file, show_col_types = FALSE)
}
```

Data cleaning is done in multiple steps. First, all radiocarbon dates that are not usable because of missing information are removed. Then the records are filtered for settlement sites, exact duplicates are removed, and duplicate determinations with identical laboratory identifiers are reduced to one.

```r
need_xronos_dedup <- refresh_xronos ||
  !file.exists(xronos_dedup_file) ||
  !file.exists(xronos_report_file)

if (need_xronos_dedup) {
  message("Preparing and deduplicating Danish XRONOS radiocarbon data...")

  chron_dk_c14_unfiltered <- chron_dk_raw |>
    mutate(bp = as.numeric(bp), std = as.numeric(std),
           site_type = as.character(site_type)) |>
    filter(!is.na(bp), !is.na(std), std > 0)

  chron_dk_c14 <- chron_dk_c14_unfiltered |>
    filter(!is.na(site_type),
           grepl(xronos_site_type_filter, site_type, ignore.case = TRUE))

  chron_dk_c14_exact <- chron_dk_c14 |>
    distinct()

  chron_dk_c14_dedup <- chron_dk_c14_exact |>
    mutate(
      dedup_key = case_when(
        !is.na(labnr) & labnr != "" ~ paste0("lab:", labnr),
        TRUE ~ paste("nolab", bp, std, site, material, species, sep = "|")
      )
    ) |>
    arrange(dedup_key) |>
    distinct(dedup_key, .keep_all = TRUE)

  dedup_report <- tibble(
    raw_records = nrow(chron_dk_raw),
    usable_c14_records_before_site_type_filter =
     ↪  nrow(chron_dk_c14_unfiltered),
    usable_c14_records_after_site_type_filter = nrow(chron_dk_c14),
    excluded_by_site_type_filter = nrow(chron_dk_c14_unfiltered) -
     ↪  nrow(chron_dk_c14),
    site_type_filter = xronos_site_type_filter,
    after_exact_deduplication = nrow(chron_dk_c14_exact),
    after_determination_deduplication = nrow(chron_dk_c14_dedup),
    removed_exact_duplicates = nrow(chron_dk_c14) - nrow(chron_dk_c14_exact),
    removed_determination_duplicates = nrow(chron_dk_c14_exact) -
     ↪  nrow(chron_dk_c14_dedup)
  )
```

```
  write_csv(chron_dk_c14_dedup, xronos_dedup_file)

  write_csv(dedup_report, xronos_report_file)
} else {
  message("Using cached deduplicated XRONOS data: ", xronos_dedup_file)

  chron_dk_c14_dedup <- read_csv(xronos_dedup_file, show_col_types = FALSE)

  dedup_report <- read_csv(xronos_report_file, show_col_types = FALSE)
}
```

Of 9,816 Danish records in XRONOS, 4,643 are usable radiocarbon dates from settlement sites. After deduplication, 996 determinations remain (Table 6).

```
dedup_steps <- tibble(
  Step = c("Danish records in XRONOS", "With usable radiocarbon age and
   ↪  error",
           "From settlement sites", "After removing exact duplicates",
           "After removing duplicate determinations (same laboratory
            ↪  number)"),
  Records = c(dedup_report$raw_records,
              dedup_report$usable_c14_records_before_site_type_filter,
              dedup_report$usable_c14_records_after_site_type_filter,
              dedup_report$after_exact_deduplication,
              dedup_report$after_determination_deduplication)
) |>
  mutate(
    Removed = c(NA, -diff(Records)),
    Records = format(Records, big.mark = ","),
    Removed = ifelse(is.na(Removed), "-", format(Removed, big.mark = ",",
     ↪  trim = TRUE))
  )

kbl(dedup_steps, align = "lrr")
```

Calibration and summed probability distribution

The deduplicated radiocarbon data are calibrated using `rcarbon` (Crema and Bevan 2021) and IntCal20 (Reimer et al. 2020). Again using `rcarbon`, we sum the resulting probability

Table 6: Selection and deduplication of the Danish XRONOS radiocarbon dates.

Step	Records	Removed
Danish records in XRONOS	9,816	–
With usable radiocarbon age and error	9,810	6
From settlement sites	4,643	5,167
After removing exact duplicates	2,632	2,011
After removing duplicate determinations (same laboratory number)	996	1,636

distribution, while binning by site name. The summed calibration is generated over a wider window than the aoristic reconstruction to avoid edge effects.

```
need_xronos_calibration <- refresh_xronos ||
  !file.exists(xronos_spd_file) ||
  !file.exists(xronos_cal_file) ||
  !file.exists(xronos_spd_rds_file)

if (need_xronos_calibration) {
  message("Calibrating Danish XRONOS radiocarbon data and building
   ↪ site-binned SPD...")

  chron_dk_c14_dedup <- chron_dk_c14_dedup |>
    mutate(
      site_bin = as.character(site),
      site_bin = if_else(is.na(site_bin) | site_bin == "",
                         paste0("missing_site_", row_number()), site_bin)
    )

  cal_dk <- calibrate(x = chron_dk_c14_dedup$bp,
                      errors = chron_dk_c14_dedup$std,
                      ids = chron_dk_c14_dedup$id,
                      calCurves = rep("intcal20", nrow(chron_dk_c14_dedup)),
                      verbose = FALSE)

  spd_dk <- spd(cal_dk, bins = chron_dk_c14_dedup$site_bin,
                timeRange = xronos_time_range_bp, spdnormalised = FALSE)

  spd_df <- tibble(cal_bp = spd_dk$grid$calBP, prd = spd_dk$grid$PrDens) |>
    mutate(year = 1950 - cal_bp)

  saveRDS(cal_dk, xronos_cal_file)
  saveRDS(spd_dk, xronos_spd_rds_file)
```

```
  write_csv(spd_df, xronos_spd_file)
} else {
  message("Using cached Danish XRONOS SPD: ", xronos_spd_file)

  spd_df <- read_csv(xronos_spd_file, show_col_types = FALSE)

  cal_dk <- readRDS(xronos_cal_file)
  spd_dk <- readRDS(xronos_spd_rds_file)
}
```

```
p_xronos_spd <- ggplot(spd_df, aes(x = year, y = prd)) +
  geom_line() +
  plot_theme(base_size = 9) +
  labs(
    title = "Summed calibrated radiocarbon probability: Denmark",
    subtitle = "XRONOS settlement dates, deduplicated and binned by site
     ↪  name",
    x = "Year BCE/CE",
    y = "Summed probability"
  )

p_xronos_spd
```

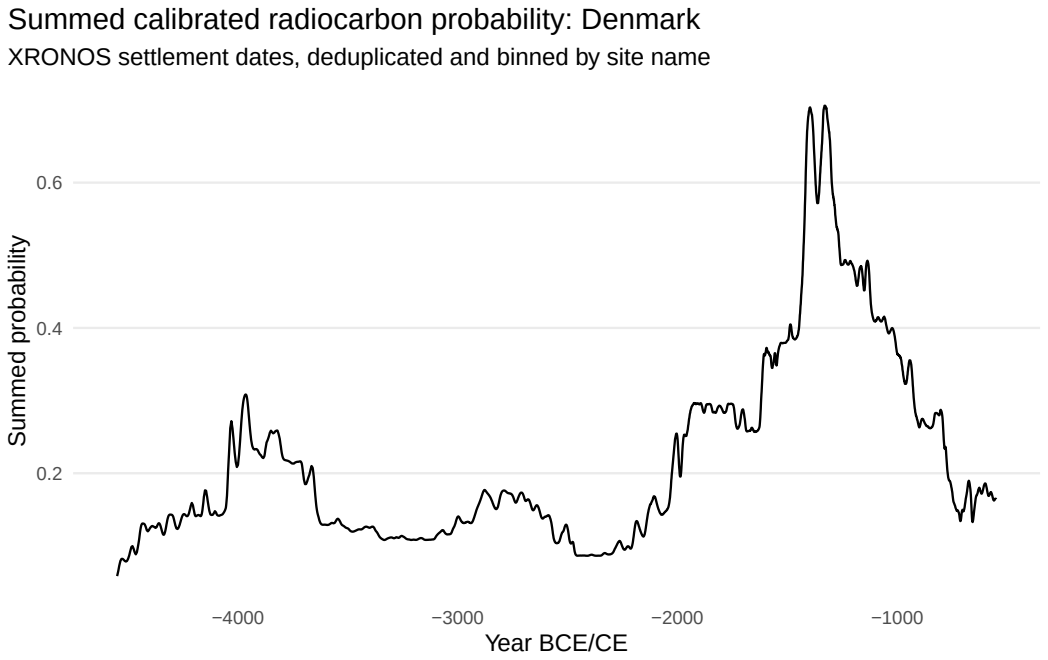


Figure 12: Summed calibrated probability distribution of the Danish XRONOS settlement dates.

Binned radiocarbon comparison

For comparison with the aoristic reconstructions, the summed calibrated radiocarbon curve is binned to the same temporal resolution and normalised to sum to one.

```
xronos_spd_binned <- spd_df |>
  filter(year >= time_min, year < time_max) |>
  mutate(year = floor((year - time_min) / bin_width) * bin_width + time_min)
   ↪  |>
  group_by(year) |>
  summarise(xronos_sumcal = sum(prd, na.rm = TRUE), .groups = "drop") |>
  mutate(xronos_sumcal = normalise_curve(xronos_sumcal))
```

```
curves_with_xronos <- curves_binned |>
  left_join(xronos_spd_binned, by = "year") |>
  mutate(xronos_sumcal = normalise_curve(replace_na(xronos_sumcal, 0)))

write_csv(curves_with_xronos,
          "outputs/s4_danish_neolithic_curves_with_xronos.csv")
```

All correlations with the SPD are computed within the common Neolithic window from 3950 to 1800 BCE. Before 3950 BCE, the most frequent start date of the phase-dated records, the SPD still contains late Mesolithic settlement dates, whereas the phase-dated dataset contains only Neolithic records. Including those bins would compare two different populations. The robustness section below also reports the full analytical window.

```
xronos_common_min <- -3950L

curves_common <- curves_with_xronos |>
  filter(year >= xronos_common_min, year < time_max)

xronos_cor <- function(col) safe_cor(curves_common$xronos_sumcal,
 ↪  curves_common[[col]])
```

Within the common window, the phase-corrected reconstruction correlates positively with the SPD at 0.645, while the standard proportional and Monte Carlo curves are essentially uncorrelated with it, at -0.050 and -0.046. The figure below shows where the curves differ. The phase-corrected curve tracks the summed calibration most closely in the sections where it departs from the standard aoristic result. Whether these correlations exceed what chance would produce is assessed in the robustness section. Baorista shows why coupling to the overlap structure and agreement with the SPD have to be reported separately. Its correlation with the overlap curve (0.319) lies between those of the standard and the phase-corrected reconstruction, but it is practically uncorrelated with the SPD (-0.132).

```
plot_xronos_comparison <- curves_with_xronos |>
  select(year, Standard_aoristic = standard, Monte_Carlo_mean = mc_mean,
         Phase_corrected = phase_corrected,
         Baorista_posterior_median = baorista_median,
         XRONOS_sumcal = xronos_sumcal) |>
  pivot_longer(cols = -year, names_to = "method", values_to = "value") |>
  mutate(
    method = recode(method, Standard_aoristic = "Standard aoristic",
                    Monte_Carlo_mean = "Monte Carlo mean",
                    Phase_corrected = "Phase-corrected",
                    Baorista_posterior_median = "Baorista posterior median",
                    XRONOS_sumcal = "XRONOS SPD"),
    method = factor(
      method,
      levels = c("Standard aoristic", "Monte Carlo mean", "Phase-corrected",
                 "Baorista posterior median", "XRONOS SPD")
    )
  )
```

```r
xronos_colours <- c(method_colours, "XRONOS SPD" = "black")

xronos_linetypes <- c(method_linetypes, "XRONOS SPD" = "dotted")

p_xronos_comparison <- ggplot(
  plot_xronos_comparison,
  aes(x = year, y = value, colour = method, linetype = method)
) +
  geom_line(linewidth = 0.8, na.rm = TRUE) +
  scale_colour_manual(values = xronos_colours, drop = FALSE) +
  scale_linetype_manual(values = xronos_linetypes, drop = FALSE) +
  coord_cartesian(xlim = c(plot_time_min, plot_time_max), ylim = c(0, y_cut))
  ↪  +
  peak_note(plot_xronos_comparison) +
  labs(
    title = "Aoristic reconstructions and Danish radiocarbon evidence",
    subtitle = paste0(
      "All curves are binned and normalised to sum to one; displayed from ",
      abs(plot_time_min), " to ", abs(plot_time_max), " BCE"
    ),
    x = "Year BCE/CE",
    y = "Normalised intensity"
  ) +
  plot_theme(base_size = 9)

p_xronos_comparison
```

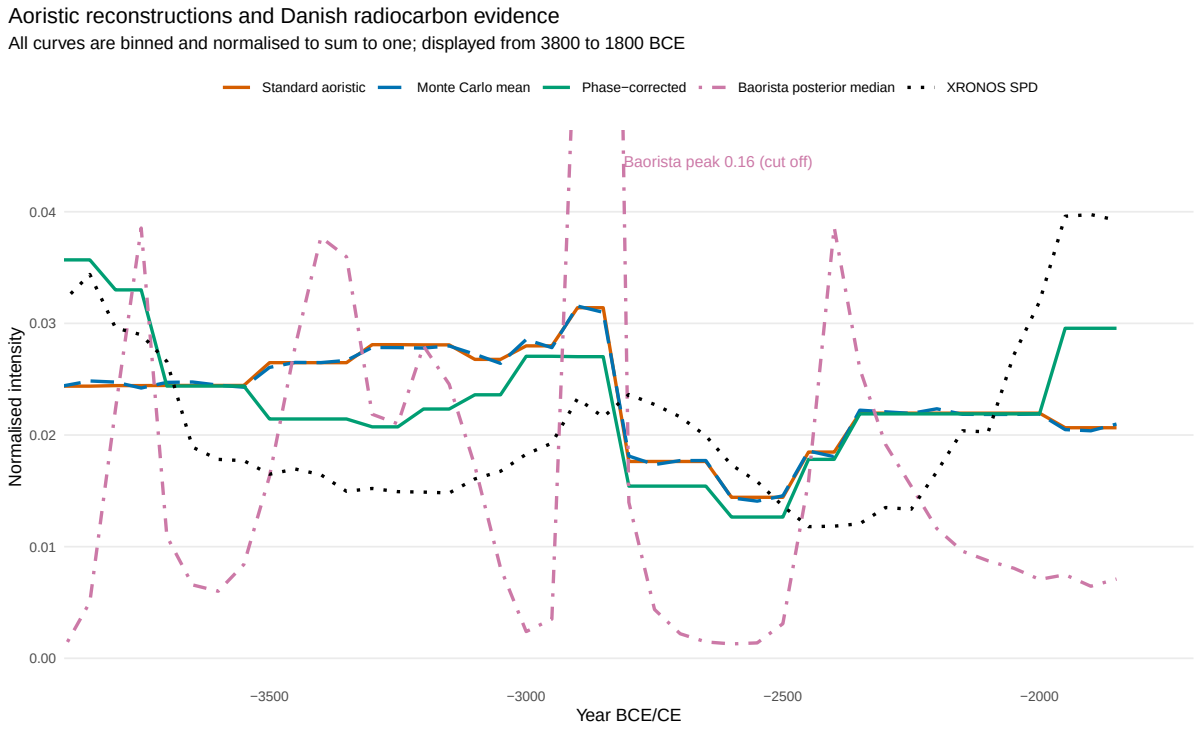


Figure 13: Aoristic reconstructions and the XRONOS SPD. The vertical axis is cut at 0.045, and the value of the Baorista peak is given in the plot.

```
ggsave("figures/s4_danish_neolithic_methods_vs_xronos_sumcal.png",
       p_xronos_comparison, width = 8.5, height = 5.2, dpi = 600)
```

```
xronos_diagnostics <- curves_common |>
  select(xronos_sumcal, standard, mc_mean, phase_corrected, baorista_median)
   ↪  |>
  summarise(standard = safe_cor(xronos_sumcal, standard),
            mc_mean = safe_cor(xronos_sumcal, mc_mean),
            phase_corrected = safe_cor(xronos_sumcal, phase_corrected),
            baorista_median = safe_cor(xronos_sumcal, baorista_median)) |>
  pivot_longer(cols = everything(), names_to = "method",
               values_to = "xronos_sumcal_correlation") |>
  mutate(
    method = recode(method, standard = "Standard aoristic",
                    mc_mean = "Monte Carlo mean",
                    phase_corrected = "Phase-corrected",
                    baorista_median = "Baorista posterior median")
  ) |>
```

Table 7: Correlation of the reconstructions with the XRONOS SPD in the common Neolithic window.

Method	Correlation with the SPD (common Neolithic window)
Phase-corrected	0.645
Baorista posterior median	-0.132
Standard aoristic	-0.050
Monte Carlo mean	-0.046

```
  mutate(window = "Common Neolithic window") |>
  arrange(desc(abs(xronos_sumcal_correlation)))

write_csv(xronos_diagnostics,
          "outputs/s4_danish_neolithic_xronos_diagnostics.csv")

xronos_diagnostics |>
  select(method, xronos_sumcal_correlation) |>
  kbl(
    digits = 3,
    col.names = c("Method", "Correlation with the SPD (common Neolithic
    ↪  window)")
  )
```

Both diagnostics point in the same direction. The phase-corrected result is the least coupled to the overlap structure and the closest to the radiocarbon evidence, although no curve agrees with the SPD beyond chance (see below). The standard reconstruction thus appears to partly reproduce the chronological classification.

The comparison also shows local offsets between the radiocarbon curve and the phase-corrected curve in parts of the sequence. These offsets point to an effect that the overlap diagnostic does not capture. Aoristic assignment is sensitive to the absolute length and placement of phases as well as to their overlap. Phases that are too long, too short, or shifted in calendar time displace probability mass even after the overlap has been corrected.

Robustness of the radiocarbon comparison

The correlations above are computed over 43 bins of 50 years. Neighbouring bins are strongly autocorrelated, above all in the SPD, so a conventional significance test would be far too optimistic. We therefore compare the observed correlations with those obtained for surrogate SPDs that keep the power spectrum, and thus the autocorrelation, of the observed SPD but have randomised Fourier phases (Theiler et al. 1992). Since the same surrogates are used for

all methods, the test can also be applied to the difference between the correlations of the phase-corrected and the standard reconstruction, which is the actual question of the comparison.

```
n_surrogates <- 10000L

phase_surrogate <- function(x) {
  n <- length(x)
  mu <- mean(x)
  f <- fft(x - mu)
  half <- 2:(floor((n - 1) / 2) + 1)
  f[half] <- Mod(f[half]) * exp(1i * runif(length(half), 0, 2 * pi))
  f[n + 2 - half] <- Conj(f[half])
  Re(fft(f, inverse = TRUE)) / n + mu
}

xronos_methods <- c(standard = "Standard aoristic",
                    mc_mean = "Monte Carlo mean",
                    phase_corrected = "Phase-corrected",
                    baorista_median = "Baorista posterior median")

xronos_windows <- list(
  "Common Neolithic window" = c(xronos_common_min, time_max),
  "Analytical window" = c(time_min, time_max)
)

set.seed(20261007)

xronos_robustness <- imap_dfr(xronos_windows, function(win, window_name) {
  d <- curves_with_xronos |>
    filter(year >= win[1], year < win[2])

  surrogates <- replicate(n_surrogates, phase_surrogate(d$xronos_sumcal))

  per_method <- imap_dfr(xronos_methods, function(label, col) {
    r <- safe_cor(d[[col]], d$xronos_sumcal)
    null <- apply(surrogates, 2, function(s) safe_cor(d[[col]], s))
    tibble(window = window_name, comparison = label, n_bins = nrow(d),
           correlation = r,
           p_surrogate = (sum(abs(null) >= abs(r)) + 1) / (n_surrogates + 1))
  })

  diff_obs <- safe_cor(d$phase_corrected, d$xronos_sumcal) -
    safe_cor(d$standard, d$xronos_sumcal)
```

Table 8: Correlation with the XRONOS SPD in the common Neolithic window and two-sided p-values against phase-randomised surrogate SPDs.

Comparison	Correlation	p (surrogates)
Standard aoristic	-0.050	0.902
Monte Carlo mean	-0.046	0.909
Phase-corrected	0.645	0.115
Baorista posterior median	-0.132	0.417
Difference: phase-corrected minus standard	0.694	0.084

```
  diff_null <- apply(surrogates, 2, function(s) {
    safe_cor(d$phase_corrected, s) - safe_cor(d$standard, s)
  })

  bind_rows(
    per_method,
    tibble(
      window = window_name,
      comparison = "Difference: phase-corrected minus standard",
      n_bins = nrow(d),
      correlation = diff_obs,
      p_surrogate = (sum(abs(diff_null) >= abs(diff_obs)) + 1) /
        ↪  (n_surrogates + 1)
    )
  )
})

write_csv(xronos_robustness,
          "outputs/s4_danish_neolithic_xronos_robustness.csv")
```

```
xronos_robustness |>
  filter(window == "Common Neolithic window") |>
  transmute(Comparison = comparison, Correlation = round(correlation, 3),
            `p (surrogates)` = round(p_surrogate, 3)) |>
  kbl()
```

None of the individual correlations is distinguishable from what an SPD of the same smoothness but random shape would produce (phase-corrected: $p = 0.115$, standard: $p = 0.902$), which is not surprising for so few effectively independent bins. The difference between the two correlations, 0.694, reaches $p = 0.084$ in the common window. Over the full analytical window, which also contains the late Mesolithic part of the SPD, it is 0.494 ($p = 0.027$). The result is

therefore a comparative one. The correction moves the reconstruction towards the radiocarbon evidence, but neither curve agrees with the SPD on its own.

Interpretation

Without a known truth, the case study shows above all that uncertainty propagation, robustness to the prior and protection against structural bias are different properties of a reconstruction.

Standard and Monte Carlo aoristics share a central tendency because they share the within-interval uniformity assumption. The phase correction changes the relative prominence of several periods. Its much lower correlation with the overlap curve, its smaller overlap-aligned amplitude and the shift towards the XRONOS SPD indicate that the standard reconstruction is partly structured by the intervals, as in the simulations.

For Baorista, the peak at 2900–2801 BCE persists across prior rates from 1 to 100 and is therefore not caused by the smoothing prior. This does not make it archaeological, because all four fits use the same interval catalogue, and the peak falls immediately before the shared boundary at 2800 BCE. The problem thus lies partly in the chronological observations and not only in the reconstruction algorithm.

The local offsets between the phase-corrected curve and the SPD show that overlap is not the only issue. Interval-based reconstructions, aoristic and Bayesian, are sensitive to the absolute duration, placement and repetition of the ranges they are given, so probability can be displaced even after overlap correction when phase definitions are too broad or imperfectly aligned with calendar chronology.

The limits of the comparison are those of any empirical test without ground truth. The case study cannot show that the corrected curve is the true settlement trajectory. The SPD depends on sampling, preservation, research history, calibration, and site-level aggregation, and neither the corrected nor the standard reconstruction agrees with it beyond chance on its own. What holds in both windows is the direction of the shift between them, towards the radiocarbon evidence. Divergence between the curves may therefore reflect biases in the typological chronology, in the radiocarbon sample, or in both.

In this dataset the standard reconstruction is strongly coupled to the overlap structure, and we therefore regard the corrected curve as the better-founded estimate. Where the two differ in parts of the sequence in which the true signal may plausibly be concentrated, both should be reported and the difference discussed. Future probabilistic frameworks may need to model the hierarchical structure, repeated assignment, shared boundaries, and overlap of archaeological phase systems in addition to the uncertainty within each interval.

Session information

```
sessionInfo()
```

```
R version 4.5.3 (2026-03-11)
Platform: aarch64-apple-darwin20
Running under: macOS 27.2

Matrix products: default
BLAS:   /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/lib
 ↪  Rblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/lib
 ↪  Rlapack.dylib;  LAPACK version 3.12.1

locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: Europe/Berlin
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices datasets  utils     methods   base

other attached packages:
 [1] rcarbon_1.5.2     xronos_0.1.1.9000 here_1.0.2        rlang_1.3.0
 [5] readr_2.1.6       tibble_3.3.1      purrr_1.2.0       ggplot2_4.0.3
 [9] tidyr_1.3.1       dplyr_1.2.1

loaded via a namespace (and not attached):
 [1] gtable_0.3.6           xfun_0.54              spatstat.sparse_3.1-0
 [4] lattice_0.22-9         aoristAAR_0.1.1        tzdb_0.5.0
 [7] Rdpack_2.6.6           vctrs_0.7.3            tools_4.5.3
[10] doSNOW_1.0.20          spatstat.utils_3.2-0   generics_0.1.4
[13] goftest_1.2-3          parallel_4.5.3         proxy_0.4-28
[16] pkgconfig_2.0.3        spatstat_3.5-0         Matrix_1.7-4
[19] KernSmooth_2.23-26     RColorBrewer_1.1-3     S7_0.2.2
[22] lifecycle_1.0.5        compiler_4.5.3         farver_2.1.2
[25] deldir_2.0-4           tinytex_0.59           spatstat.linnet_3.4-0
[28] codetools_0.2-20       spatstat.explore_3.6-0 snow_0.4-4
[31] htmltools_0.5.9        class_7.3-23           yaml_2.3.12
[34] crayon_1.5.3           pillar_1.11.1          spatstat.model_3.5-0
[37] classInt_0.4-11        spatstat.univar_3.1-5  iterators_1.0.14
```

```
[40] rpart_4.1.24           abind_1.4-8             foreach_1.5.2
[43] nlme_3.1-168           spatstat.geom_3.6-1     tidyselect_1.2.1
[46] digest_0.6.39          sf_1.0-23               labeling_0.4.3
[49] splines_4.5.3          polyclip_1.10-7         rprojroot_2.1.1
[52] fastmap_1.2.0          grid_4.5.3              cli_3.6.6
[55] magrittr_2.0.5         e1071_1.7-16            spatstat.data_3.1-9
[58] withr_3.0.3            tensor_1.5.1            scales_1.4.0
[61] bit64_4.6.0-1          rmarkdown_2.30          bit_4.6.0
[64] hms_1.1.4              evaluate_1.0.5          knitr_1.50
[67] rbibutils_2.4.1        mgcv_1.9-4              Rcpp_1.1.0
[70] spatstat.random_3.4-3  glue_1.8.1              DBI_1.2.3
[73] renv_1.2.0             vroom_1.6.7             jsonlite_2.0.0
[76] R6_2.6.1               units_1.0-0
```

References

Crema, ER. 2025 A Bayesian alternative for aoristic analyses in archaeology. *Archaeometry* 67(S1): 7–30. DOI: https://doi.org/10.1111/arcm.12984.

Crema, ER and Bevan, A. 2021 Inference from large sets of radiocarbon dates: Software and methods. *Radiocarbon* 63(1): 23–39. DOI: https://doi.org/10.1017/RDC.2020.95.

Reimer, PJ, Austin, WEN, Bard, E, Bayliss, A, Blackwell, PG, Bronk Ramsey, C, Butzin, M, Cheng, H, Edwards, RL, Friedrich, M, Grootes, PM, Guilderson, TP, Hajdas, I, Heaton, TJ, Hogg, AG, Hughen, KA, Kromer, B, Manning, SW, Muscheler, R, Palmer, JG, Pearson, C, van der Plicht, J, Reimer, RW, Richards, DA, Scott, EM, Southon, JR, Turney, CSM, Wacker, L, Adolphi, F, Büntgen, U, Capano, M, Fahrni, SM, Fogtmann-Schulz, A, Friedrich, R, Köhler, P, Kudsk, S, Miyake, F, Olsen, J, Reinig, F, Sakamoto, M, Sookdeo, A and Talamo, S. 2020 The IntCal20 Northern Hemisphere radiocarbon age calibration curve (0–55 cal kBP). *Radiocarbon* 62(4): 725–757. DOI: https://doi.org/10.1017/RDC.2020.41.

Roe, J, Schmid, C, Ebrahimiabareghi, S, Heitz, C and Hinz, M. 2025 XRONOS: An open data infrastructure for archaeological chronology. *Journal of Computer Applications in Archaeology* 8(1): 242–263. DOI: https://doi.org/10.5334/jcaa.191.

Slots- og Kulturstyrelsen. 2018. *Fund og fortidsminder*. 2018. Available at https://www.kulturarv.dk/fundogfortidsminder/.

Theiler, J, Eubank, S, Longtin, A, Galdrikian, B and Farmer, JD. 1992 Testing for nonlinearity in time series: The method of surrogate data. *Physica D: Nonlinear Phenomena* 58(1–4): 77–94. DOI: https://doi.org/10.1016/0167-2789(92)90102-S.

Turnbull, BW. 1976 The empirical distribution function with arbitrarily grouped, censored and truncated data. *Journal of the Royal Statistical Society. Series B (Methodological)* 38(3): 290–295. DOI: https://doi.org/10.1111/j.2517-6161.1976.tb01597.x.